Adaptive Dilation Factor Design Strategies for the Implicit Feature Extraction Module in SSSDTCN-based Industrial Time Series Imputation

Technical Research Plan

Abstract. The SSSDTCN model uses a dilated causal convolution stack as its implicit feature extraction module, with dilation rates fixed to the exponential schedule $\{1, 2, 4, 8, 16, 32\}$ before training. While this hand-crafted design achieves broad multi-scale temporal coverage, it cannot adapt to the specific periodicity or correlation structure of different industrial datasets or input segments. This document surveys and formalises **five principled approaches** for making the dilation factors *adaptive*: (I) learnable soft dilation parameters, (II) input-conditioned selective kernel fusion (SKNet-inspired), (III) differentiable neural architecture search (DARTS-based), (IV) frequency-spectral guided dilation, and (V) temporal self-attention guided dynamic dilation. For each approach the core adaptive mechanism, its source literature, and the essential mathematical formulation are presented together with an illustrative diagram.

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Background

The implicit module of SSSDTCN stacks L dilated causal convolution layers. A 1-D dilated causal convolution with kernel $\mathbf{w}$, kernel size k, and dilation d computes:

$$(\mathbf{x} \star_d \mathbf{w})[t] = \sum_{i=0}^{k-1} \mathbf{w}[i] \cdot \mathbf{x}[t - d \cdot i], \tag{1}$$

enforcing causality via left-only zero-padding. The current model fixes $d_\ell = 2^{\ell-1}$, giving rates $\{1, 2, 4, 8, 16, 32\}$ for $L = 6$ and a total receptive field of $\mathrm{RF} = 1+(k-1)(2^L-1) = 253$ steps (for $k = 5$). Three limitations motivate adaptive designs: (1) the Debutanizer and SRU datasets have distinct dominant periodicities that a single fixed schedule cannot simultaneously optimise; (2) under high missing-ratio regimes longer context matters more, but the schedule is mask-blind; (3) non-stationary operating regimes cannot be tracked by static rates. Figure 1 illustrates the current fixed-dilation stack.

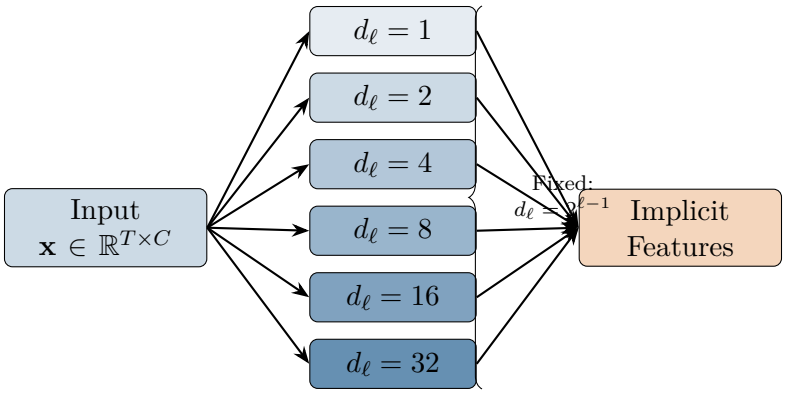

Figure 1: Current SSSDTCN implicit module: six dilated causal convolution layers with a fixed exponential dilation schedule.

1 Approach I: Learnable Soft Dilation Parameters

Inspired by **Deformable Convolutional Networks** [Dai et al., 2017], which showed that sampling offsets in convolutional kernels can be learned from data rather than fixed on a regular grid, this approach extends the same principle to the temporal dilation axis. Each layer ℓ is assigned an unconstrained trainable scalar $\lambda_\ell \in \mathbb{R}$; the effective dilation factor is $d_\ell = \exp(\lambda_\ell)$, which is always positive and differentiable. For non-integer d_ℓ, the convolution uses linear interpolation between adjacent time steps, so the gradient flows back to λ_ℓ via the chain rule. The key to adaptivity is that $\{\lambda_\ell\}$ are jointly optimised with all other model weights during end-to-end training, so the dilation schedule converges to the values that best explain the training data—rather than the values a human engineer pre-selected.

Core equations.

$$d_\ell = \exp(\lambda_\ell), \tag{2}$$

$$(\mathbf{x} \star_{d_\ell} \mathbf{w})[t] = \sum_{i=0}^{k-1} \mathbf{w}[i] \cdot \underbrace{(1 - \{d_\ell i\})\, \mathbf{x}[\lfloor t - d_\ell i \rfloor] + \{d_\ell i\}\, \mathbf{x}[\lceil t - d_\ell i \rceil]}_{\text{linear interpolation}}, \tag{3}$$

$$\mathcal{L} = \mathcal{L}_{\text{impute}} - \beta \sum_{\ell < \ell'} |d_\ell - d_{\ell'}|, \tag{4}$$

where $\{u\}$ is the fractional part of u and the diversity term eq. (4) (weight $\beta > 0$) prevents all layers from collapsing to the same dilation value.

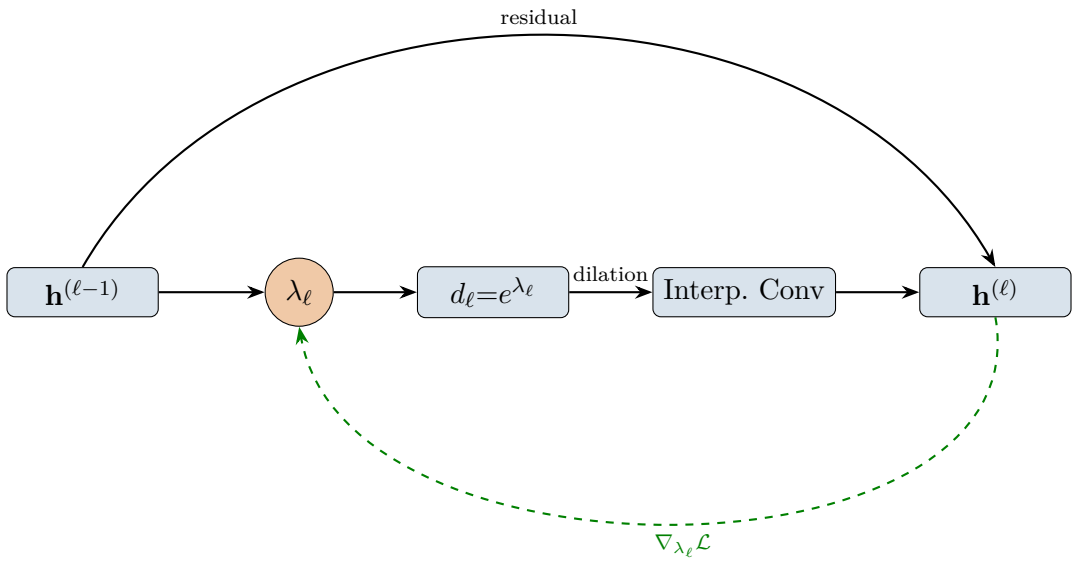

Figure 2: Approach I: the dilation $d_\ell = e^{\lambda_\ell}$ is a trainable scalar. Gradients (green dashed) flow through the interpolated convolution and update λ_ℓ end-to-end.

2 Approach II: Input-Conditioned Selective Kernel Dilation

This approach adapts the idea of **Selective Kernel Networks (SKNet)** [Li et al., 2019] and **Squeeze-and-Excitation Networks (SENet)** [Hu et al., 2018] to the temporal dilation setting. SKNet showed that, for a given input, a network can dynamically choose which kernel size to emphasise by routing features through a lightweight attention module. Here, K parallel dilated convolutions with distinct fixed candidate rates $D = \{d^{(1)}, \ldots, d^{(K)}\}$ are applied to the same input simultaneously. Their outputs are aggregated, globally average-pooled into a compact descriptor, and passed through two fully-connected layers to produce per-channel soft selection weights $\alpha^{(j)} \in [0,1]^H$ for each branch. The final output is the weighted sum of all branch outputs. Adaptivity arises because the selection weights are *conditioned on the current input*: different input segments or samples yield different $\alpha^{(j)}$, so the effective dilation shifts automatically to match the dominant temporal scale of each input.

Core equations.

$$\tilde{\mathbf{h}}^{(j)} = \mathrm{BN}\Big(\mathrm{Conv}_{d^{(j)}}(\mathbf{h}^{(\ell-1)})\Big), \quad j = 1, \ldots, K, \tag{5}$$

$$\mathbf{s} = \mathrm{AvgPool}\Big(\textstyle\sum_j \tilde{\mathbf{h}}^{(j)}\Big) \in \mathbb{R}^H, \tag{6}$$

$$\boldsymbol{\alpha}^{(j)} = \mathrm{softmax}_j\Big(\mathbf{W}_2^{(j)}\mathrm{ReLU}(\mathbf{W}_1\mathbf{s})\Big) \in [0,1]^H, \tag{7}$$

$$\mathbf{h}^{(\ell)} = \sum_{j=1}^{K} \boldsymbol{\alpha}^{(j)} \odot \tilde{\mathbf{h}}^{(j)}. \tag{8}$$

The input-level effective dilation at time t is interpretable as $d_\ell^{\mathrm{eff}}(t) = \sum_j \bar{\alpha}^{(j)}(t) \cdot d^{(j)}$, where $\bar{\alpha}^{(j)}(t)$ is the channel-mean weight.

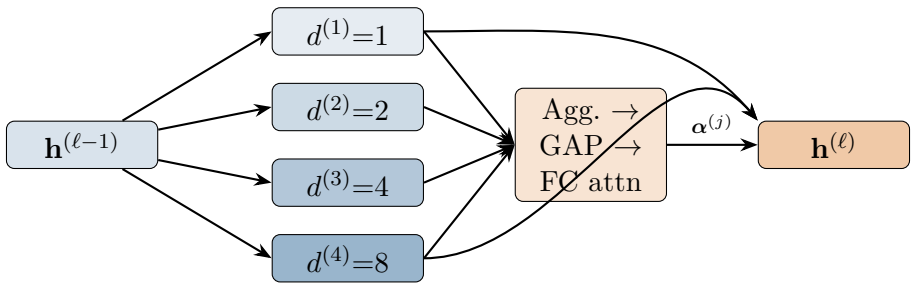

Figure 3: Approach II: four parallel branches with different dilation rates. The squeeze-and-excitation block produces input-conditioned per-channel attention weights that recombine the branch outputs.

3 Approach III: Differentiable Architecture Search for Dilation Rates

DARTS (Differentiable ARchiTecture Search) [Liu et al., 2019] introduced continuous relaxation of discrete architecture choices via a *mixed operation*: instead of committing to one operation, a weighted mixture of all candidate operations is used during search, with mixture weights jointly optimised via gradient descent. Translated to the dilation setting, each layer ℓ maintains a real-valued architecture parameter vector $\boldsymbol{\alpha}_\ell \in \mathbb{R}^K$ over K candidate dilations. During the search phase, the layer computes a softmax-weighted mixture of all K dilated convolutions; after search, the candidate with the highest weight is selected and the final model is retrained from scratch with those discrete rates. Adaptivity is dataset-level: the architecture search explores a rich dilation space $D = \{1, 2, 3, 4, 6, 8, 12, 16, 24, 32\}$ and converges to the schedule that minimises validation loss—no human choice required.

Core equations.

$$\bar{o}^{(\ell)}(\mathbf{h}) = \sum_{j=1}^{K} \frac{\exp(\alpha_\ell^{(j)})}{\sum_{j'} \exp(\alpha_\ell^{(j')})} \cdot \mathrm{Conv}_{d^{(j)}}(\mathbf{h}), \tag{9}$$

$$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta_\theta \nabla_{\boldsymbol{\theta}} \mathcal{L}_{\text{train}}(\boldsymbol{\theta}, \boldsymbol{\alpha}), \tag{10}$$

$$\boldsymbol{\alpha} \leftarrow \boldsymbol{\alpha} - \eta_\alpha \nabla_{\boldsymbol{\alpha}} \mathcal{L}_{\text{val}}(\boldsymbol{\theta}, \boldsymbol{\alpha}), \tag{11}$$

$$d_\ell^* = d^{(\hat{j}_\ell)}, \quad \hat{j}_\ell = \arg\max_j \alpha_\ell^{(j)}. \tag{12}$$

Equations eqs. (10) and (11) are the alternating gradient steps of the bi-level optimisation; eq. (12) is the final discretisation.

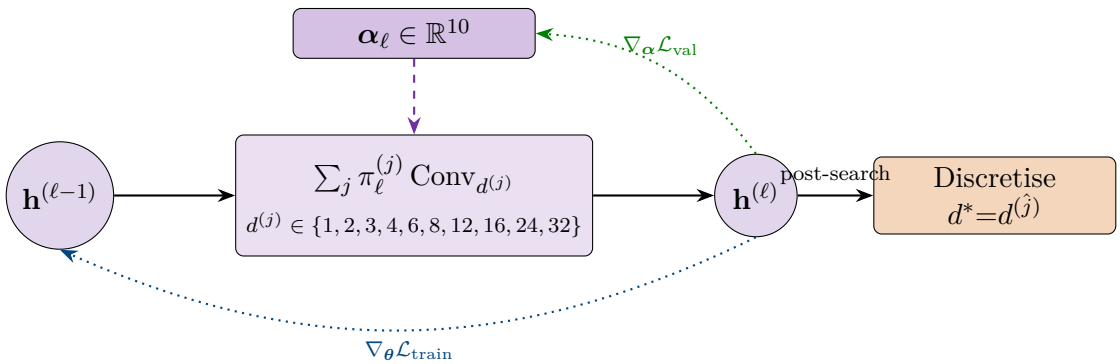

Figure 4: Approach III: DARTS mixed operation. During search all candidate dilations are active (weighted by $\pi_\ell^{(j)}$). After search the best-scoring dilation is selected (orange box).

4 Approach IV: Frequency-Spectral Guided Adaptive Dilation

TimesNet [Wu et al., 2023] and **FEDformer** [Zhou et al., 2022] demonstrated that industrial and meteorological time series carry strong periodicity, and that aligning model structure with dominant periods substantially improves performance. This approach applies the same insight directly to dilation-rate setting: the Fast Fourier Transform is applied to the training data to locate the L dominant frequency components, each is converted to a period $P_\ell = T/f_\ell$, and the dilation for layer ℓ is then set so that the kernel's sampling stride matches one full period—$d_\ell = \lfloor P_\ell/(k-1) \rfloor$. No additional trainable parameters are introduced; the entire calibration runs once as an offline pre-processing step, and the resulting dilation schedule is automatically customised to each dataset's physical periodicity.

Core equations.

$$A_f = \frac{1}{C}\sum_{c=1}^{C}\Big|\mathrm{FFT}(\mathbf{x}[c,:])[f]\Big|, \quad f = 1, \ldots, \lfloor T/2 \rfloor, \tag{13}$$

$$\{f_1, \ldots, f_L\} = \mathrm{TopK}(\{A_f\},\, L)\,, \tag{14}$$

$$P_\ell = \lfloor T/f_\ell \rceil\,, \tag{15}$$

$$d_\ell = \max(1,\, \lfloor P_\ell/(k-1) \rceil)\,. \tag{16}$$

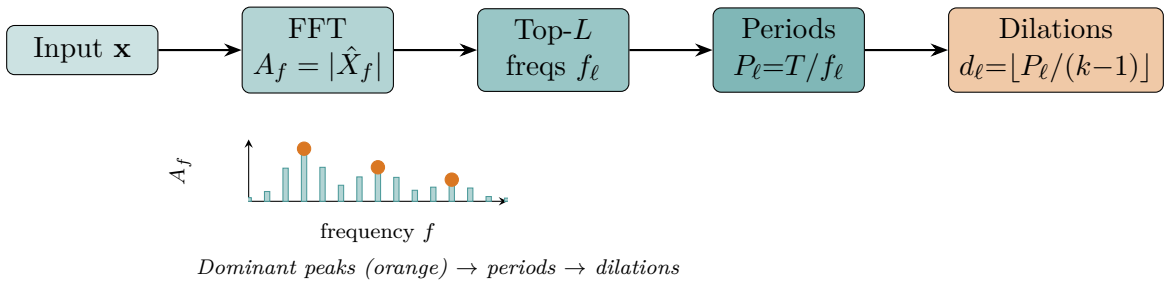

Figure 5: Approach IV: the FFT magnitude spectrum is computed once from the training set; the L largest peaks (orange dots) determine periods P_ℓ, which are mapped to dilation rates d_ℓ.

5 Approach V: Temporal Self-Attention Guided Dynamic Dilation

Dynamic Convolution [Chen et al., 2020] showed that conditioning convolution kernels on the current input context—rather than using fixed weights—substantially increases representational capacity. **Non-local Means** [Wang et al., 2018] similarly showed the benefit of using position-level attention to identify the most informative temporal offsets. Combining these ideas, this approach inserts a lightweight temporal self-attention module before the convolution: query and key projections are applied to the hidden state, producing a per-position attention score $a_{t,\delta}$ for each candidate offset δ. The expected offset $\bar{d}_t = \sum_\delta \delta\, a_{t,\delta}$ gives a continuous, position-level effective dilation that varies *within a single sequence*: high-frequency regions concentrate weight on small offsets (fine scale), while smoother or longer-range regions concentrate weight on large offsets (coarse scale). A multiplicative gate then modulates the base dilation $d_\ell^{(0)}$ of the original layer by this context-dependent factor.

Core equations.

$$e_{t,\delta} = \frac{\mathbf{q}_t^\top \mathbf{k}_{t-\delta}}{\sqrt{d_k}}, \quad a_{t,\delta} = \frac{\exp(e_{t,\delta})}{\sum_{\delta'} \exp(e_{t,\delta'})}, \tag{17}$$

$$\bar{d}_t = \sum_{\delta=0}^{T_{\max}} \delta \cdot a_{t,\delta}, \tag{18}$$

$$\tilde{d}_{t,\ell} = d_\ell^{(0)} \cdot 2\ \mathrm{sigmoid}\Big(\mathbf{w}_s^\top \mathbf{h}^{(\ell-1)}[:,t] + b_s\Big), \tag{19}$$

where eq. (19) bounds the adapted dilation in $[0, 2d_\ell^{(0)}]$.

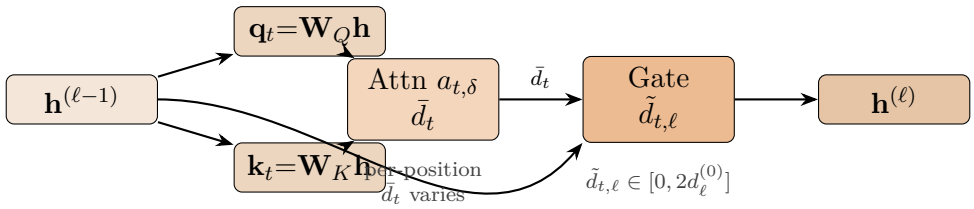

Figure 6: Approach V: a self-attention module computes a per-position expected offset $\bar{d}_t$; a sigmoid gate (right) scales the base dilation $d_\ell^{(0)}$ by a context-dependent factor, producing a fully position-adaptive effective dilation.

6 Comparative Discussion

Table 1 summarises the five approaches along key design dimensions. The recommended starting point is a hybrid of Approaches IV and I: use the spectral calibration (eq. (16)) to set the initial $\lambda_\ell^{(0)} = \log(d_\ell^{\text{spectral}})$, then fine-tune with learnable soft dilations. This combines physics-guided initialisation with gradient-based refinement at minimal implementation cost.

Table 1: Comparison of adaptive dilation approaches.

Approach	**Extra params**	**Adapt. level**	**Search?**	**Infer. cost**	**Key reference**
Fixed (baseline)	0	—	No	$\mathcal{O}(LC)$	—
I. Soft dilation	L scalars	Global	No	$\mathcal{O}(LC)$	Dai et al. [2017]
II. SKNet fusion	$O(KH^2)$	Per-sample	No	$\mathcal{O}(KLC)$	Li et al. [2019]
III. DARTS	LK (search)	Global	Yes	$\mathcal{O}(LC)$	Liu et al. [2019]
IV. Spectral	0	Global	No	$\mathcal{O}(LC)$	Wu et al. [2023]
V. Attn-guided	$O(Hd_kL)$	Per-position	No	$\mathcal{O}(LCT)$	Chen et al. [2020]

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